

DATA SCIENCE COURSEWORK

Introduction to Data Science | DS201.3 | Classification Coursework

Predicting Student Stress Level Based on Social Media Usage and Lifestyle Factors

A machine learning classification study using KNN, Logistic Regression and SVM

EDA

Classification

Model Evaluation

Stress Prediction

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Prepared for a classification coursework task involving dataset selection, exploratory data analysis, model development and performance evaluation.

Report Structure

This report follows the main steps of a machine learning classification project. It includes the introduction, dataset description, exploratory data analysis, data preprocessing, model development, model evaluation, results interpretation, prediction example, conclusion and references. This structure helps to present the coursework in a clear and organised way.

Section	Description
1. Introduction	Gives a short background about the project and explains the aim of the coursework.
2. Dataset Description	Describes the selected dataset, features and target variable.
3. Exploratory Data Analysis	Shows charts and explains the main patterns found in the dataset.
4. Data Preprocessing	Explains how the dataset was cleaned and prepared before model training.
5. Model Development	Explains the classification models used in the project.
6. Model Evaluation	Compares the models using accuracy, classification reports and confusion matrices.
7. Results and Interpretation	Explains the main results and the best performing model.
8. Prediction on New Student Data	Shows how the trained model was used to predict stress level for a new student.
9. Conclusion	Summarises the overall work and final outcome of the project.
10. References	Lists the dataset source and any other references used.

1. Introduction

Social media is now a normal part of student life. Most students use social media platforms for entertainment, education, communication, networking and news. At the same time, students also have different daily habits such as study time, sleep time, physical activity and mobile phone usage. These factors can be connected with a student's stress level.

The main purpose of this coursework is to build a machine learning classification model to predict student stress level. For this project, the selected dataset is based on student social media usage and mental health impact. The dataset includes details such as age, gender, country, academic level, most used social media platform, purpose of use, average daily usage hours, daily unlocks, study hours, physical activity hours, sleep hours and mental health score.

The target variable used in this project is Stress_Level. This variable has four categories: Low, Medium, High and Very High. Since the target variable is categorical, this project is suitable for a classification task.

In this coursework, the dataset was first explored to understand the data. After that, data cleaning and preprocessing were done before building the models. Three machine learning algorithms were used for classification: K-Nearest Neighbours, Logistic Regression and Support Vector Machine. Finally, the models were evaluated and compared to select the best performing model.

2. Dataset Description

The dataset used for this coursework is Student Social Media and Mental Health Impact. This dataset contains information about students, their social media usage, daily habits and mental health related details. It is suitable for this project because the data can be used to predict a student's stress level.

The main target variable in this dataset is Stress_Level. This column shows the stress level of each student. It has four categories: Low, Medium, High and Very High. Since the target variable contains categories, this project becomes a classification problem.

The dataset includes several input features that can be useful for prediction. Some features describe the student, such as age, gender, country and academic level. Some features describe social media behaviour, such as the most used platform, purpose of use, average daily usage hours and daily phone unlocks. Other features describe lifestyle habits, such as study hours, physical activity hours, sleep hours per night and mental health score.

These features are important because student stress can be connected with both online behaviour and daily routines. For example, high social media usage, fewer sleeping hours, low physical activity or lower study balance may show some relationship with stress level. Therefore, this dataset gives enough information to explore patterns and build a classification model.

Column Name	Simple Description
Age	Age of the student
Gender	Gender of the student
Country	Country of the student
Academic_Level	Academic level of the student
Most_Used_Platform	Social media platform mostly used
Purpose_Of_Use	Main reason for using social media
Avg_Daily_Usage_Hours	Average hours spent on social media per day
Daily_Unlocks	Number of times the phone is unlocked daily
Study_Hours	Time spent studying
Physical_Activity_Hours	Time spent on physical activities
Sleep_Hours_Per_Night	Average sleep hours per night
Mental_Health_Score	Mental health score of the student
Stress_Level	Target variable to predict

This dataset is suitable for the coursework because it has a clear target variable and useful input features. It also supports the main steps required in the coursework, such as data exploration, visualisation, preprocessing, model training and model evaluation.

3. Exploratory Data Analysis

Exploratory Data Analysis was done to understand the dataset before building the machine learning models. In this step, the main patterns, distributions and relationships in the data were observed using charts. This helped to understand how social media usage and lifestyle factors are connected with student stress levels.

3.1 Stress Level Distribution

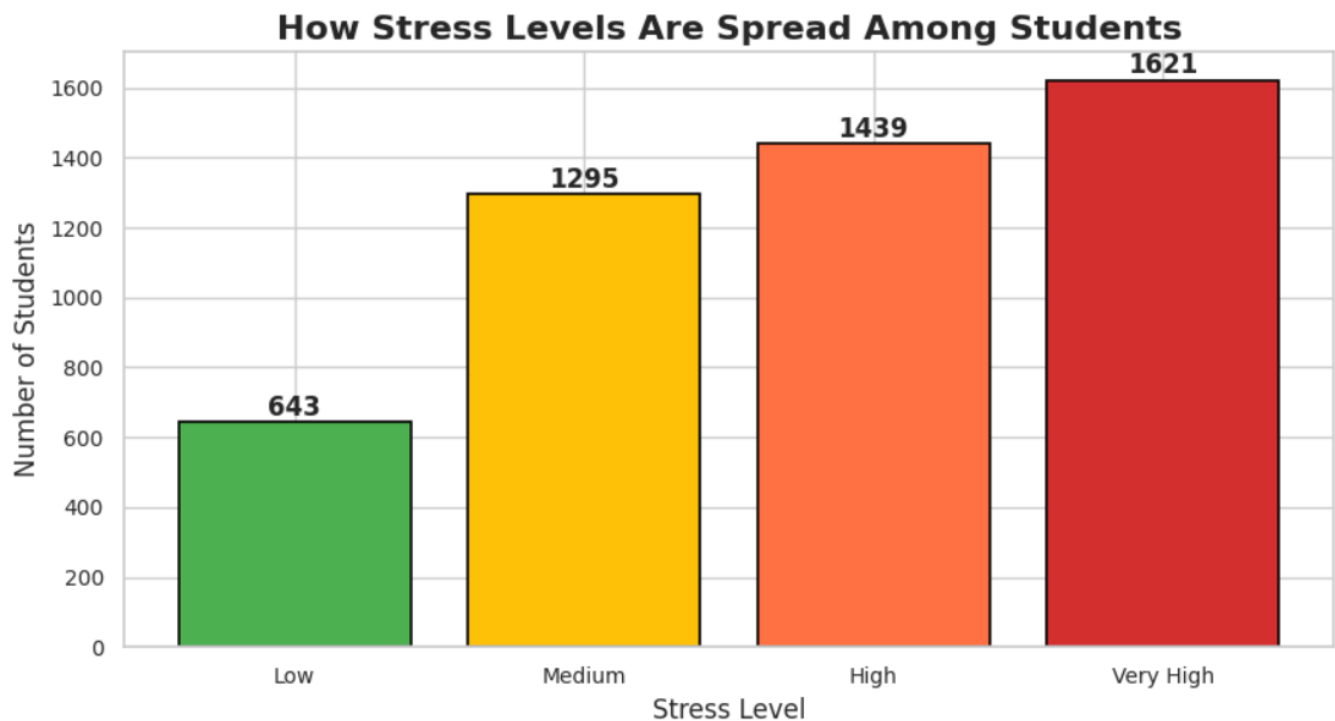


Figure 1: Distribution of Stress Levels

This chart shows how students are distributed across different stress levels. It helps to understand whether the target classes are balanced or not. This is important because the model learns from these stress level categories.

3.2 Percentage View of Stress Levels

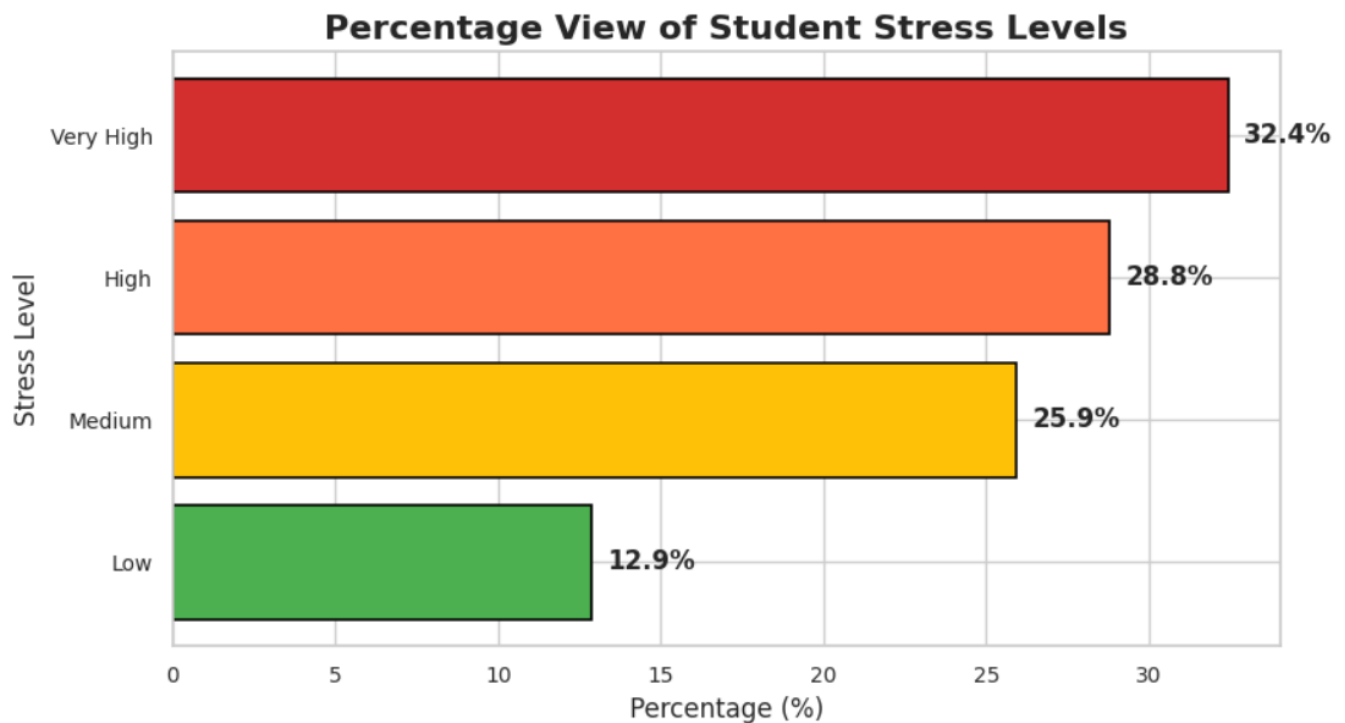


Figure 2: Percentage View of Stress Levels

This chart shows the percentage of students in each stress level category. It gives a clearer idea about the class distribution. If one class has a much higher percentage, it can affect the model performance.

3.3 Social Media Usage and Stress Level

This chart compares daily social media usage hours with stress levels. It helps to see whether students with higher usage hours show higher stress levels. This is useful because social media usage is one of the main features in the dataset.

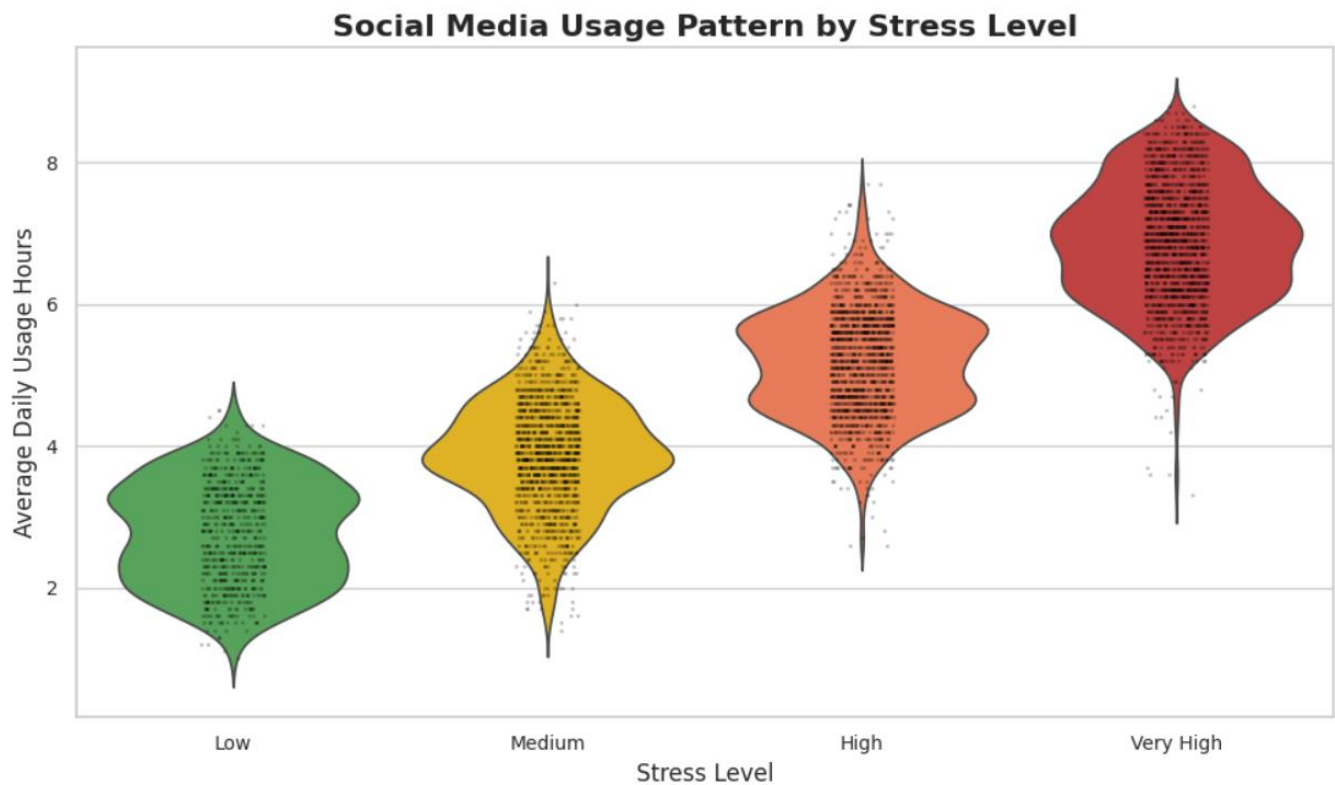


Figure 3: Social Media Usage Pattern by Stress Level

3.4 Sleep Hours and Stress Level

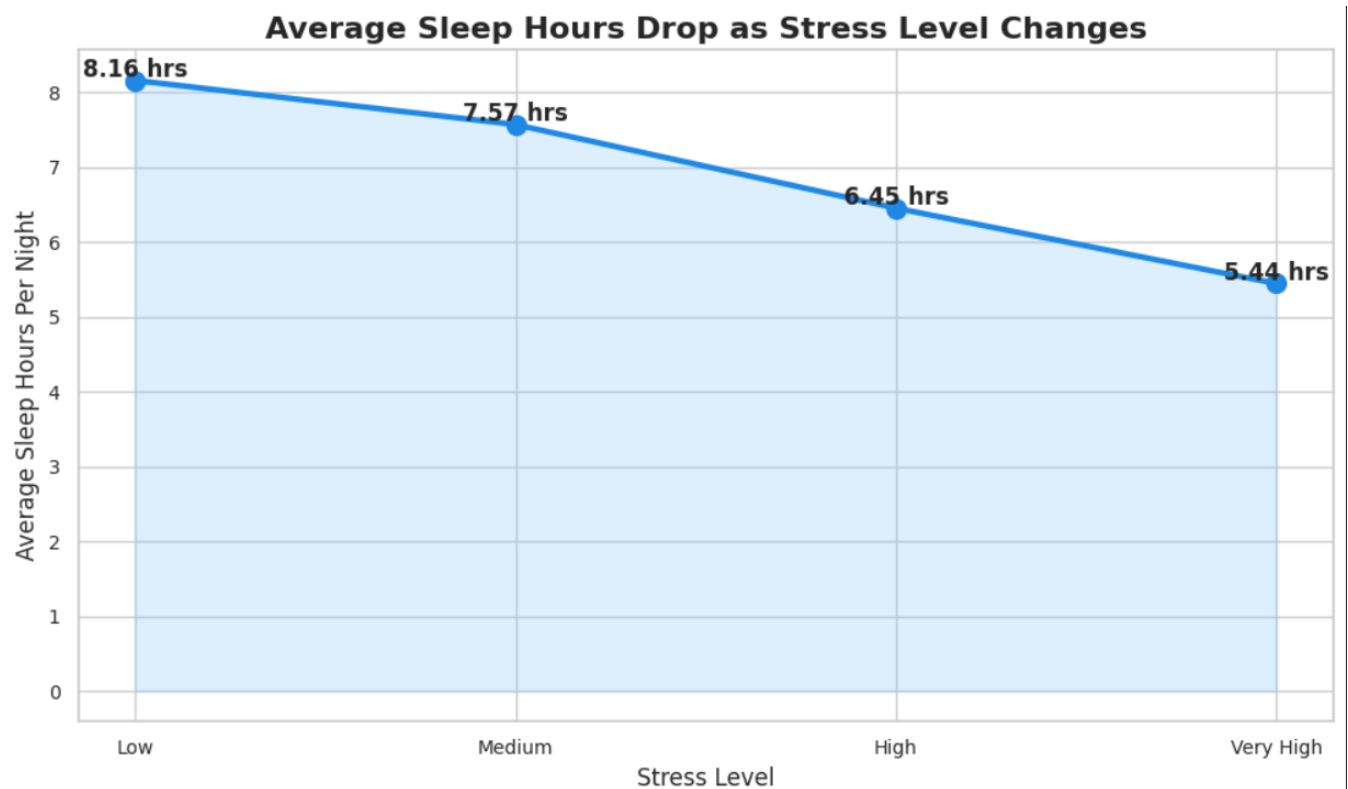


Figure 4: Average Sleep Hours by Stress Level

This chart shows the average sleeping hours for each stress level. Sleep is an important daily habit, so this chart helps to check whether students with different stress levels have different sleep patterns.

3.5 Study Hours and Stress Level

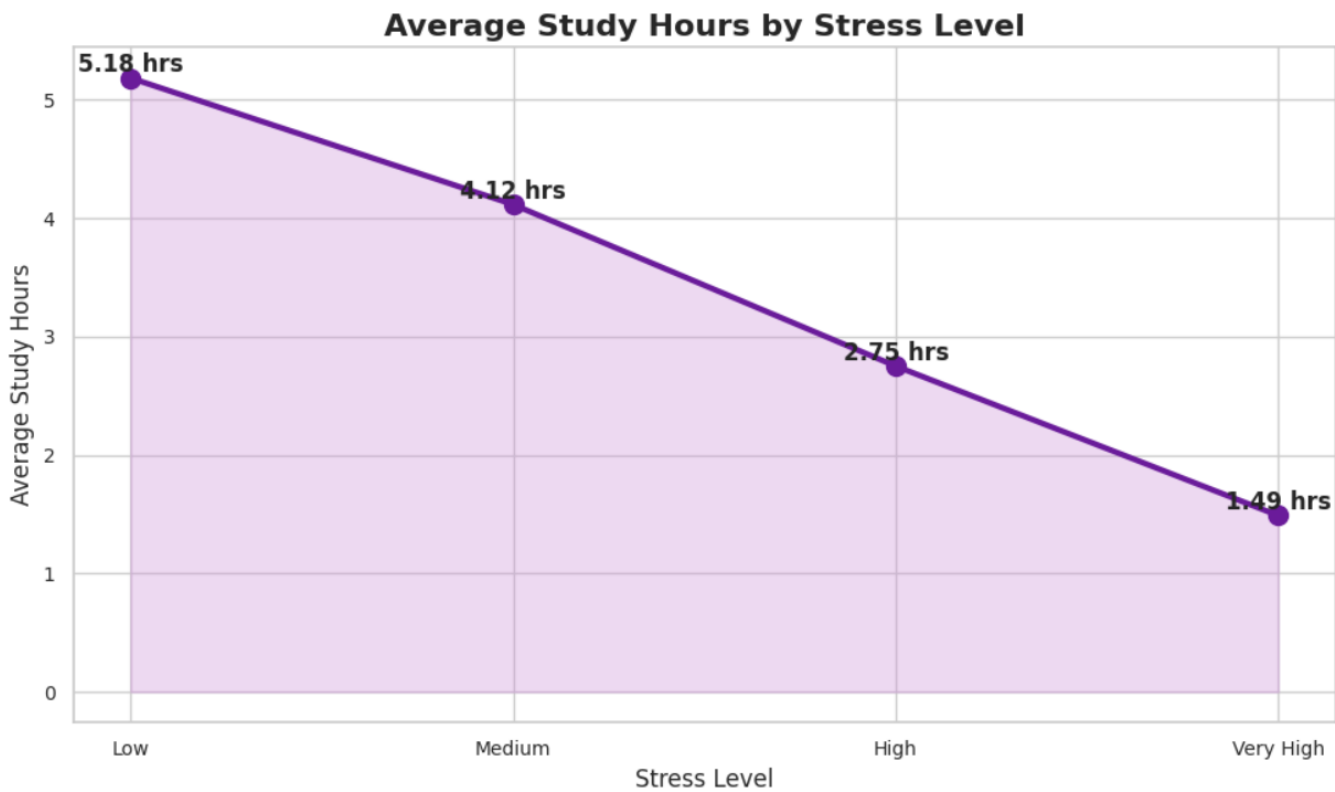


Figure 5: Average Study Hours by Stress Level

This chart compares study hours across stress levels. It helps to understand whether study time changes according to the stress level of students.

3.6 Physical Activity and Stress Level

This chart shows the average physical activity hours for each stress level. Physical activity is a lifestyle factor, so it is useful to check whether it has any visible pattern with student stress levels.

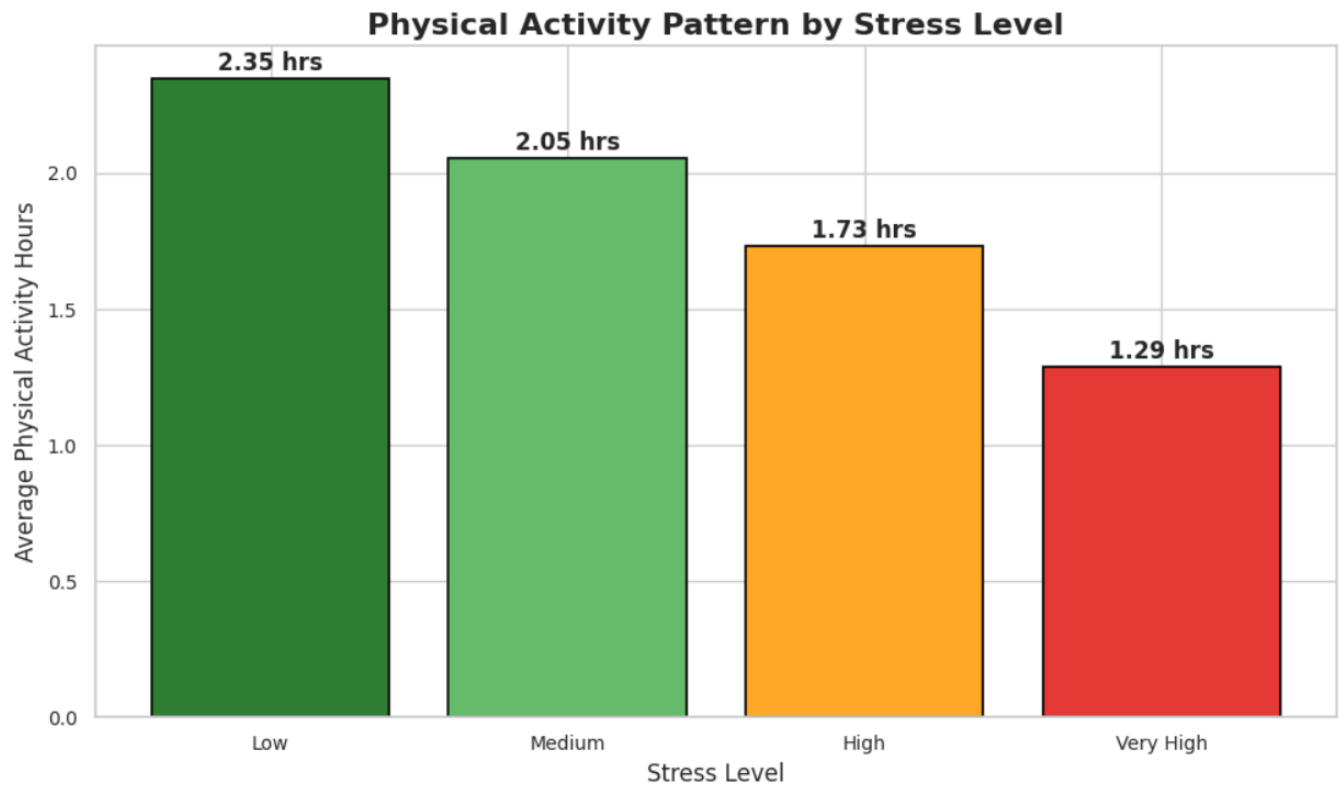


Figure 6: Physical Activity Pattern by Stress Level

3.7 Platform and Stress Level

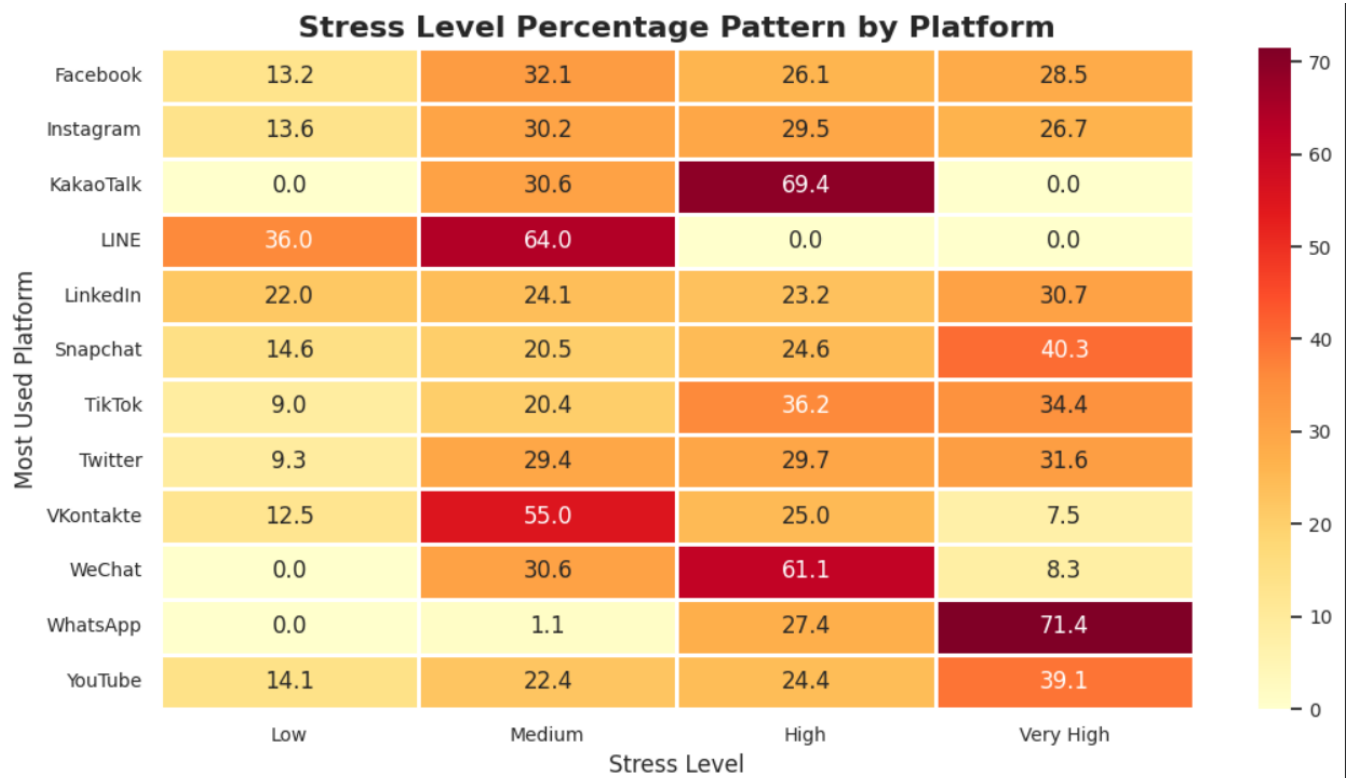


Figure 7: Stress Level Percentage Pattern by Platform

This heatmap shows how stress levels are distributed across different social media platforms. It makes it easier to compare stress patterns between platform groups.

3.8 Correlation Between Numerical Features

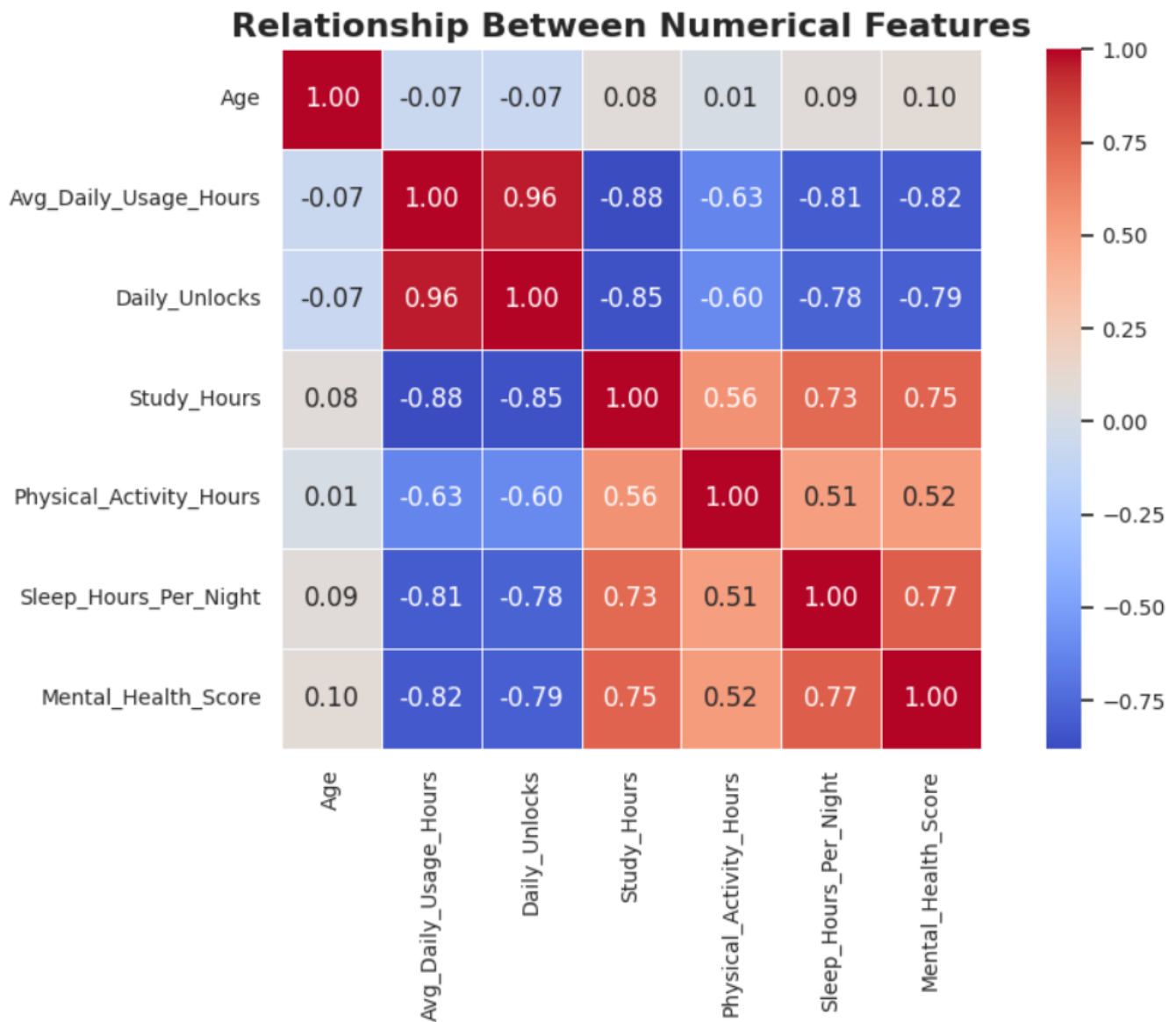


Figure 8: Correlation Heatmap of Numerical Features

The correlation heatmap shows the relationship between numerical features in the dataset. It helps to identify which features move together and which features may be useful for model training.

Overall, the EDA helped to understand the main features in the dataset. The charts showed patterns related to stress level, social media usage, sleep, study time, physical activity and platform usage. These observations were useful before moving to data preprocessing and model development.

4. Data Preprocessing

Data preprocessing was done before training the machine learning models. This step is important because raw data may contain missing values, duplicate records and categorical values that cannot be directly used by machine learning algorithms.

First, a copy of the original dataset was created so that the original data would not be changed. Then the column names were cleaned to remove any unwanted spaces. After that, duplicate rows were checked and removed from the dataset.

Missing values were also checked. For numerical columns, missing values were handled using the median value. For categorical columns, missing values were handled using the most common value. This helped to keep the dataset clean and ready for model training.

The target variable used in this project was `Stress_Level`. The remaining columns were used as input features. Since some columns contained text values, categorical features were converted into numerical form using one-hot encoding. The target variable was converted into numerical labels using label encoding.

After encoding, the dataset was split into training and testing sets. The training set was used to train the models, and the testing set was used to evaluate model performance. Feature scaling was also applied because the selected models such as KNN, Logistic Regression and SVM can be affected by different feature scales.

Step	Purpose
Data copy	To protect the original dataset
Column name cleaning	To remove unwanted spaces in column names
Duplicate removal	To avoid repeated records
Missing value handling	To fill empty values in the dataset
Feature and target split	To separate input columns and target column
Label encoding	To convert Stress_Level into numerical labels
One-hot encoding	To convert categorical input features into numerical form
Train-test split	To divide data for training and testing
Feature scaling	To make numerical values more suitable for model training

After completing these preprocessing steps, the dataset was ready to be used for machine learning model development. This helped to make the training process more accurate and organised.

5. Model Development

After data preprocessing, machine learning models were developed to predict the stress level of students. The target variable used for prediction was Stress_Level. The input features included student background details, social media usage details and lifestyle-related details.

For this coursework, three classification algorithms were used: K-Nearest Neighbours, Logistic Regression and Support Vector Machine. These models were selected because they are suitable for classification problems and were covered during the module.

K-Nearest Neighbours predicts the class of a new record by looking at similar records in the training data. Logistic Regression predicts the probability of each class and selects the most suitable class. Support Vector Machine tries to separate the classes using a decision boundary.

Before training the models, the categorical values were converted into numerical form. Feature scaling was also applied because KNN, Logistic Regression and SVM can be affected by different value ranges in the dataset. After that, the models were trained using the training dataset.

Model	Reason for Using
K-Nearest Neighbours	Simple classification model that predicts based on similar records
Logistic Regression	Useful for classification and easy to interpret
Support Vector Machine	Useful for separating classes using a decision boundary

The trained models were later tested using the testing dataset. Their performance was compared to identify which model predicted student stress levels more accurately.

6. Model Evaluation

After training the models, their performance was evaluated using the testing dataset. Model evaluation is important because it shows how well each model can predict the stress level of students. In this project, accuracy score, classification report and confusion matrix were used to evaluate the models.

Three models were compared in this section: K-Nearest Neighbours, Logistic Regression and Support Vector Machine. The accuracy score was used to get a basic idea of how many predictions were correct. The model with the highest accuracy was selected as the best performing model.

Model	Accuracy
K-Nearest Neighbours	0.759
Logistic Regression	0.751
Support Vector Machine	0.745

According to the accuracy results, the models showed different performance levels. The accuracy values helped to compare the models and identify which model performed better for predicting student stress level.

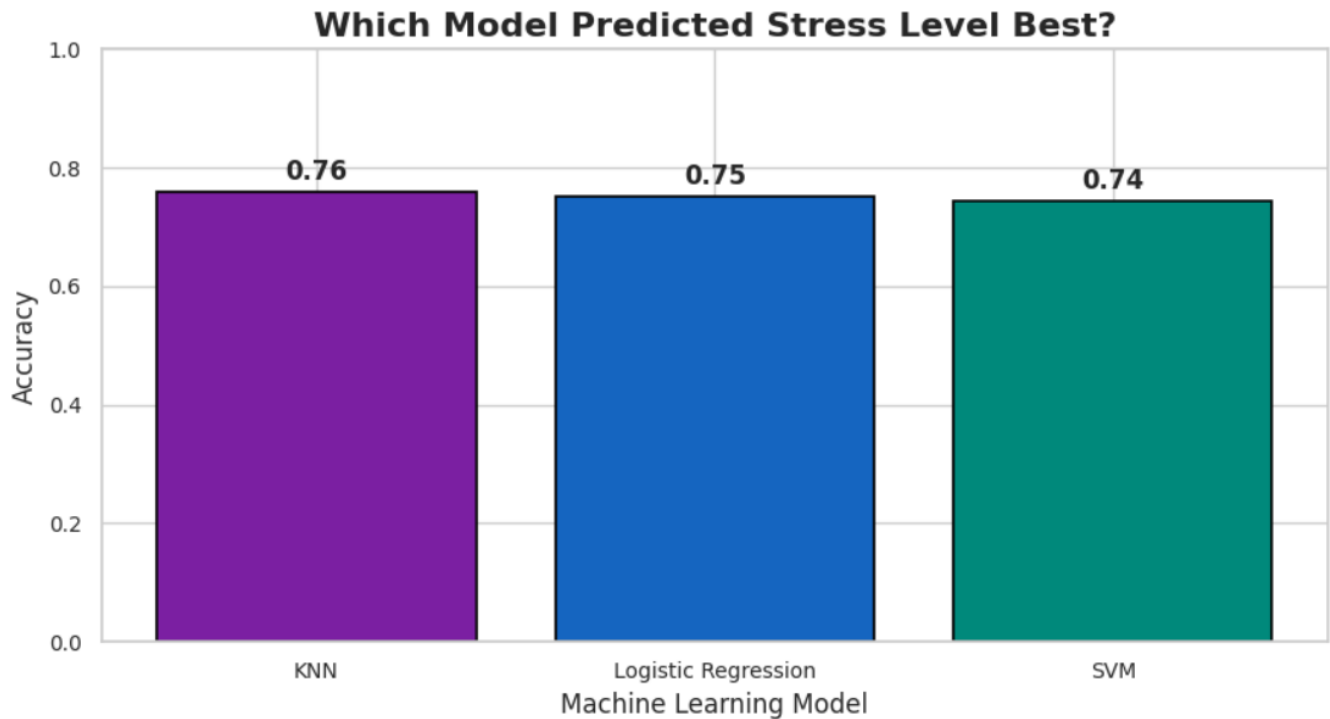


Figure 9: Accuracy Comparison of Machine Learning Models

This chart compares the accuracy of the three models. It makes it easier to see which model performed better in predicting the student stress level.

6.1 Classification Report

The classification report was used to get a more detailed idea about model performance. It shows how well each model predicted the different stress level categories. In this project, the report included precision, recall and F1-score values for each stress level class.

Precision shows how many predicted values were actually correct. Recall shows how many actual values were correctly identified by the model. F1-score gives a balanced score using both precision and recall. These values are useful because accuracy alone may not show the full performance of a classification model.

By checking the classification reports of KNN, Logistic Regression and SVM, the performance of each model was compared more clearly. This helped to understand which model handled the stress level categories better.

Metric	Meaning
Precision	How many predicted answers were correct
Recall	How many actual answers were correctly found
F1-score	A balanced score using precision and recall
Accuracy	Overall correct predictions

6.2 Confusion Matrix

A confusion matrix was used to check how well each model predicted the stress level classes. It compares the actual stress level values with the predicted values given by the model.

In this project, confusion matrices were created for K-Nearest Neighbours, Logistic Regression and Support Vector Machine. These matrices helped to identify how many predictions were correct and how many were wrong for each stress level category.

The diagonal values in a confusion matrix show the correct predictions. The other values show the incorrect predictions. Therefore, this visualisation is useful to understand the model performance more clearly than accuracy alone.

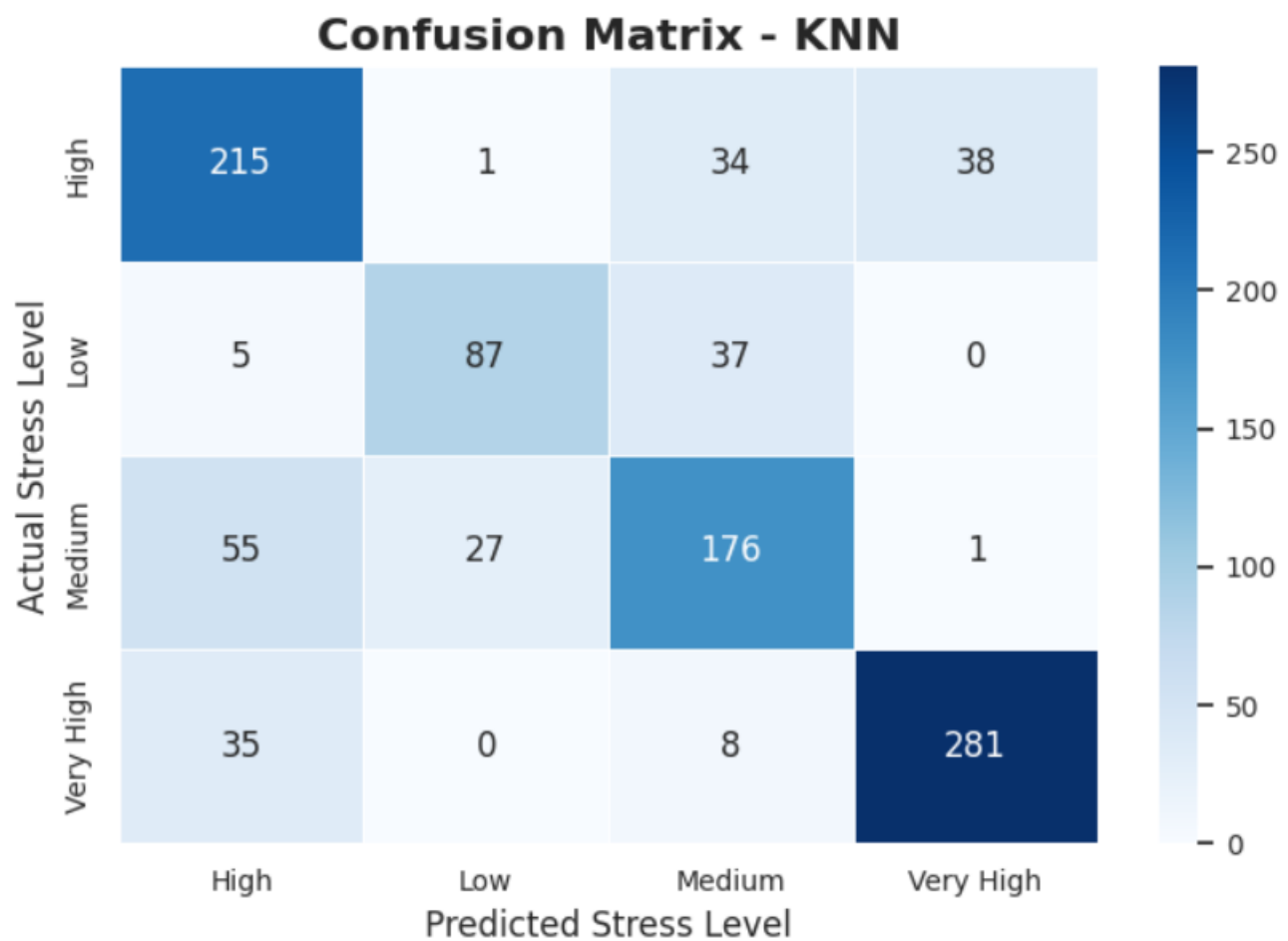


Figure 10: Confusion Matrix for K-Nearest Neighbour

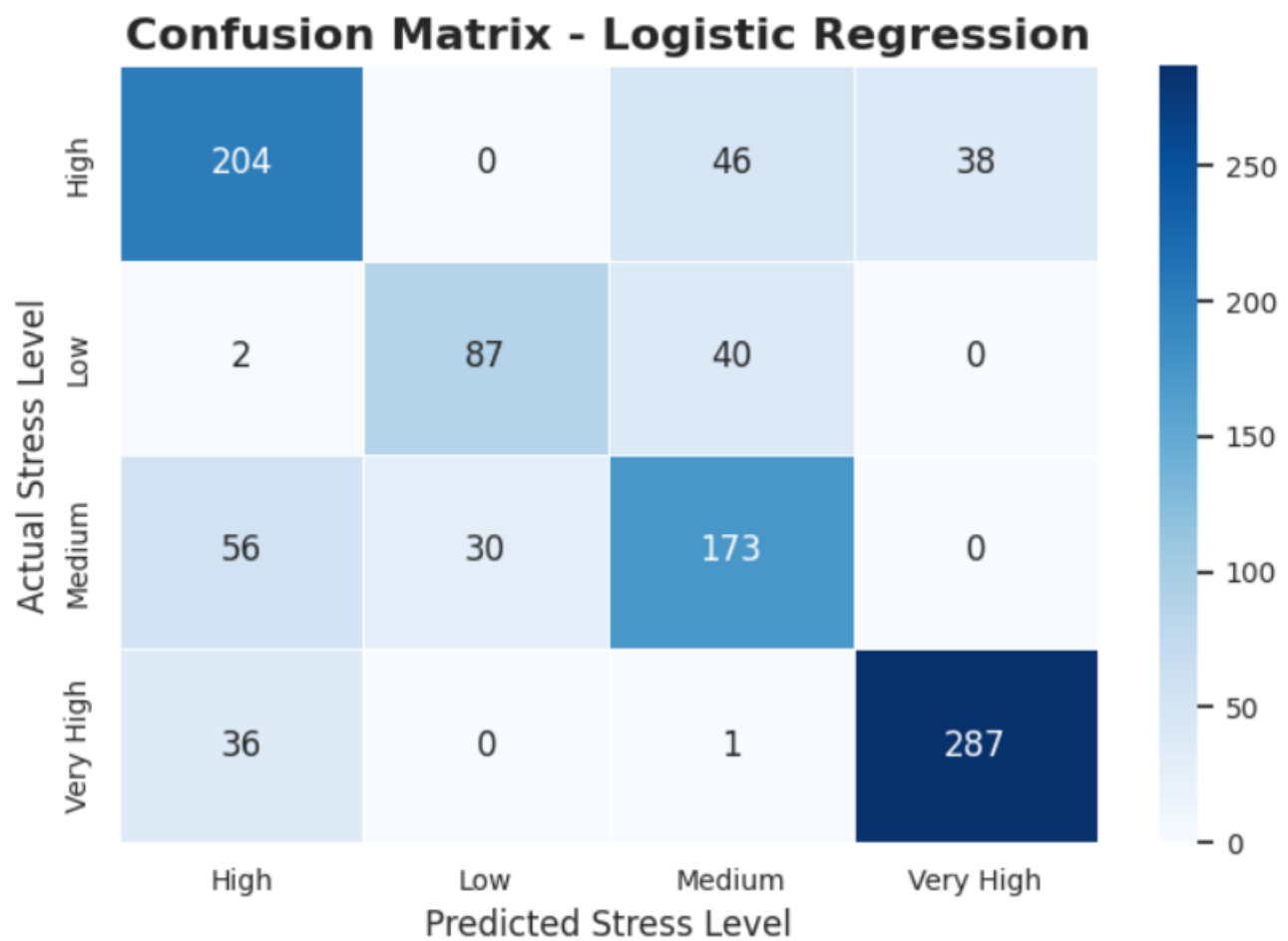


Figure 11: Confusion Matrix for Logistic Regression

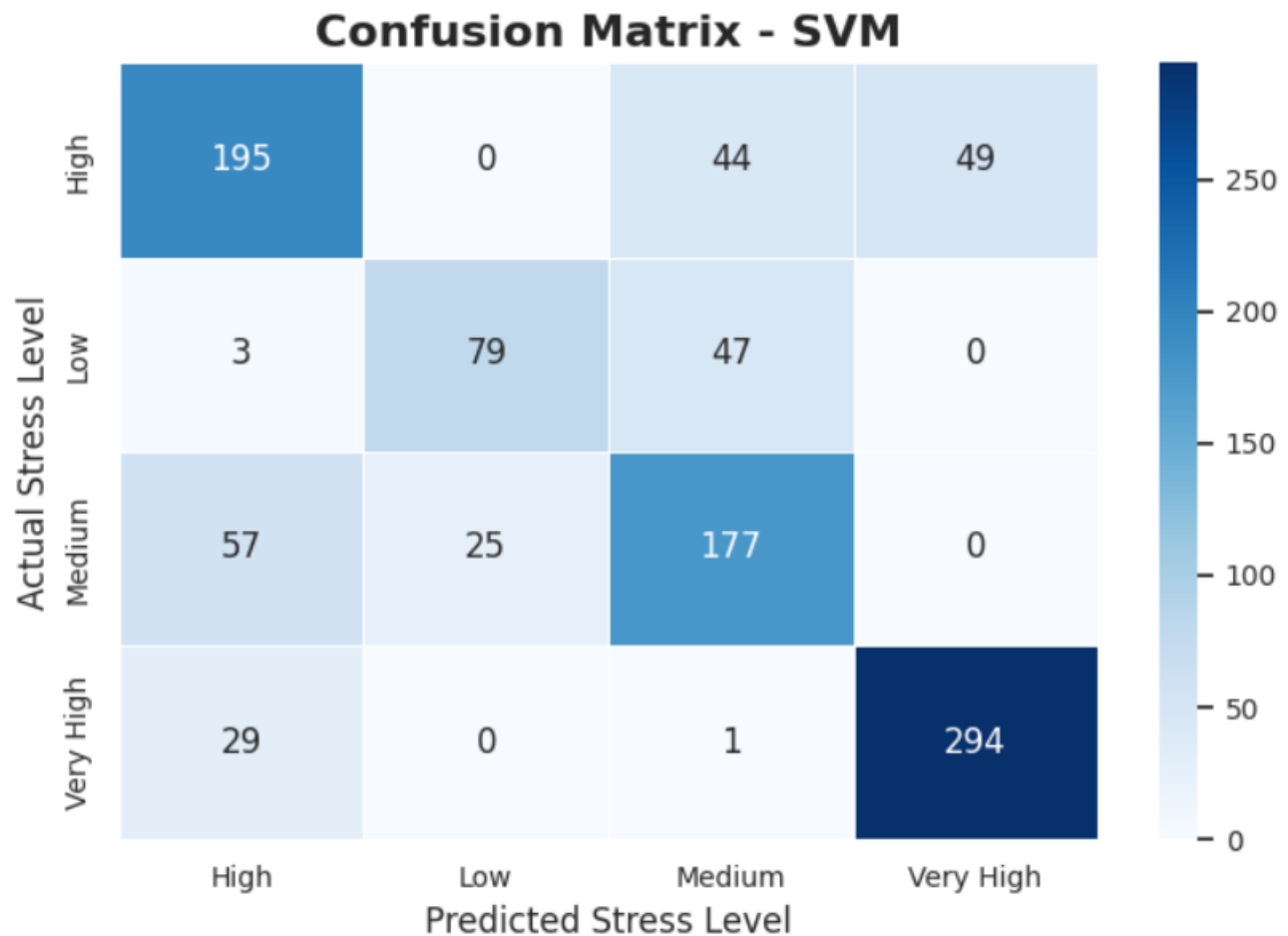


Figure 12: Confusion Matrix for Support Vector Machine

By comparing the confusion matrices, it was possible to see which model predicted the stress level categories better. A model with more values on the diagonal and fewer values outside the diagonal can be considered better. This helped to support the final model selection.

6.3 Best Model Selection

After comparing the three models, the model with the highest accuracy was selected as the best model. The best model was also checked using the classification report and confusion matrix. This helped to make the final selection more reliable.

The selected best model was used later to predict the stress level of a new student record.

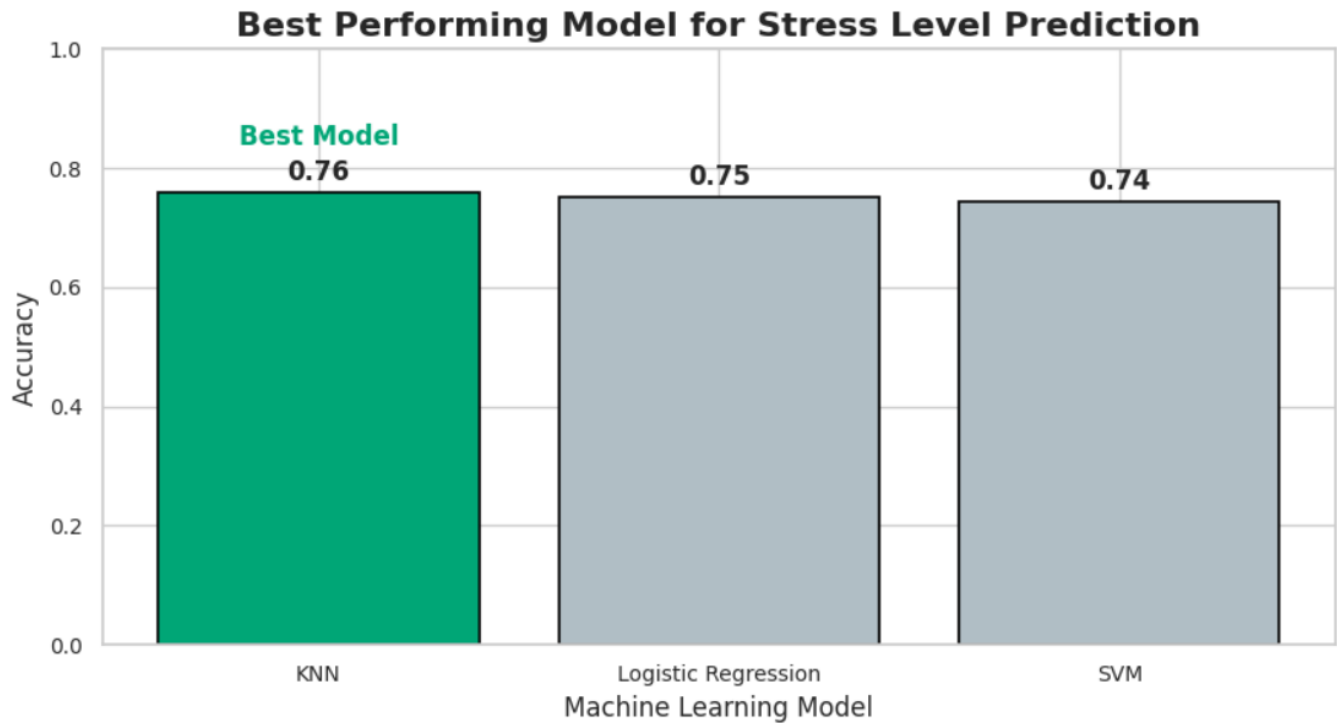


Figure 13: Best Performing Model for Stress Level Prediction

This chart highlights the model that achieved the highest accuracy. It gives a clear visual summary of the model comparison and shows which algorithm performed best for this dataset.

7. Results and Interpretation

The results show that all three models were able to predict student stress levels using the selected features. The models used in this project were K-Nearest Neighbours, Logistic Regression and Support Vector Machine. Their performance was compared using accuracy, classification reports and confusion matrices.

According to the accuracy results, K-Nearest Neighbours gave the highest accuracy among the three models. KNN achieved an accuracy of 0.759, while Logistic Regression achieved 0.751 and SVM achieved 0.745. Since KNN had the highest accuracy, it was selected as the best performing model for this dataset.

The difference between the model accuracy values was not very large, but KNN performed slightly better than the other two models. This means that, for this dataset, predicting stress level based on similar student records worked well.

The confusion matrices also helped to understand the model results more clearly. A good model should have more correct predictions on the diagonal of the confusion matrix. By looking at the confusion matrices, it was possible to compare how well each model predicted the Low, Medium, High and Very High stress level categories.

The EDA charts also supported the model results. Features such as average daily social media usage, daily phone unlocks, sleep hours, study hours, physical activity hours and mental health score showed useful patterns with stress level. These patterns helped the models to learn relationships between student behaviour and stress level.

Overall, the results show that social media usage and lifestyle-related features can be used to predict student stress level. Based on the evaluation results, KNN was selected as the final model for this coursework.

Item	Result
Target Variable	Stress_Level
Models Used	KNN, Logistic Regression, SVM
Best Model	K-Nearest Neighbours
Best Accuracy	0.759

8. Prediction on New Student Data

After selecting the best model, a new student record was created to test the prediction process. The purpose of this step was to check whether the trained model could predict the stress level for a new input record.

The new student data included details such as age, gender, country, academic level, most used social media platform, purpose of use, daily social media usage hours, daily unlocks, study hours, physical activity hours, sleep hours and mental health score.

Before making the prediction, the new student data was prepared using the same preprocessing steps used for the training data. The categorical values were encoded, and the feature values were scaled. This was done to make sure the new input matched the format used by the trained model.

Since K-Nearest Neighbours was selected as the best model, it was used to predict the stress level of the new student record. The model gave the predicted stress level based on the patterns it learned from the training data.

Feature	Value
Age	21
Gender	Male
Country	Sri Lanka
Academic Level	Undergraduate
Most Used Platform	Instagram
Purpose of Use	Entertainment
Average Daily Usage Hours	5.5
Daily Unlocks	180
Study Hours	2.5
Physical Activity Hours	1.2
Sleep Hours Per Night	6.0
Mental Health Score	6.0

Using the trained KNN model, the stress level was predicted for the new student record.

Item	Result
Final Model Used	K-Nearest Neighbours
Predicted Stress Level	High

This prediction step shows how the trained model can be used with new student data. It also shows that the same preprocessing steps must be followed before giving new data to the model.

9. Conclusion

This coursework focused on predicting student stress level using social media usage and lifestyle related factors. The selected dataset was suitable for a classification task because it had a clear target variable named Stress_Level.

Before building the models, the dataset was explored using different charts. These charts helped to understand patterns related to stress level, social media usage, sleep hours, study hours, physical activity and mental health score. Data cleaning and preprocessing were also done to prepare the dataset for model training.

Three machine learning models were used in this project: K-Nearest Neighbours, Logistic Regression and Support Vector Machine. After comparing the model accuracy values, K-Nearest Neighbours gave the highest accuracy of 0.759. Therefore, KNN was selected as the best performing model for this dataset.

Overall, this project shows that student stress level can be predicted using social media behaviour and lifestyle features. The final model can be used to predict the stress level of a new student record after applying the same preprocessing steps.

This coursework helped to understand how data science and machine learning can be applied to a real-world student related problem.

10. References

The dataset used in this coursework was collected from a publicly available online source. It was used for educational purposes to perform exploratory data analysis and build machine learning classification models.

Dataset:

Student Social Media and Mental Health Impact Dataset. Available at:

<https://www.kaggle.com/datasets/shivasingh4945/student-social-media-and-mental-health-impact>

Software and Tools:

Google Colab was used to write and run the Python code. Python libraries such as Pandas, Matplotlib, Seaborn and Scikit-learn were used for data analysis, visualisation, preprocessing, model training and model evaluation.

Source	Description
Student Social Media and Mental Health Impact Dataset	Dataset used for the classification task
Google Colab	Platform used for coding and model training
Pandas	Used for data handling and cleaning
Matplotlib and Seaborn	Used for data visualization
Scikit-learn	Used for preprocessing, model training and evaluation